

Chapter 4, Section 2

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1. Use (2.5.3) to show that $\mathbb{P}^n$ is simply connected.

Proof. Proceed by induction on n . The base case $n = 1$ is (2.5.3). Let $f : X \rightarrow \mathbb{P}^n$ be an étale covering. We may assume X is connected. Then X is smooth over k since f is étale (III, 10.1), and X is projective over k since f is finite, so X is a smooth irreducible projective variety of dimension n . It is enough to show the degree of f is 1. By (III, Ex. 11.3), $f^{-1}(H) = X \times_{\mathbb{P}^n} H$ is smooth and irreducible for almost every hyperplane $H \subset \mathbb{P}^n$. Also, the induced morphism $f' : f^{-1}(H) \rightarrow H$ is an étale covering of $H \cong \mathbb{P}^{n-1}$, since étale morphisms are stable under base change. By the inductive hypothesis, f' is an isomorphism. Hence, f is an isomorphism. $\square$

2. *Classification of Curves of Genus 2.* Fix an algebraically closed field k of characteristic $\neq 2$.

- (a) If X is a curve of genus 2 over k , the canonical linear system $|K|$ determines a finite morphism $f : X \rightarrow \mathbb{P}^1$ of degree 2 (Ex. 1.7). Show that it is ramified at exactly 6 points, with ramification index 2 at each one. Note that f is uniquely determined, up to an automorphism of $\mathbb{P}^1$, so X determines an (unordered) set of 6 points of $\mathbb{P}^1$, up to an automorphism of $\mathbb{P}^1$.
- (b) Conversely, given six distinct elements $\alpha_1, \dots, \alpha_6 \in k$, let K be the extension of $k(x)$ determined by the equation $z^2 = (x - \alpha_1) \cdots (x - \alpha_6)$. Let $f : X \rightarrow \mathbb{P}^1$ be the corresponding morphism of curves. Show that $g(X) = 2$, the map f is the same as the one determined by the canonical linear system, and f is ramified over the six points $x = \alpha_i$ of $\mathbb{P}^1$, and nowhere else.
- (c) Using (I, Ex. 6.6), show that if P_1, P_2, P_3 are three distinct points of $\mathbb{P}^1$, then there exists a unique $\varphi \in \text{Aut } \mathbb{P}^1$ such that $\varphi(P_1) = 0, \varphi(P_2) = 1, \varphi(P_3) = \infty$. Thus in (a), if we order the six points of $\mathbb{P}^1$, and then normalize by sending the first three to $0, 1, \infty$, respectively, we may assume that X is ramified over $0, 1, \infty, \beta_1, \beta_2, \beta_3$, where $\beta_1, \beta_2, \beta_3$ are three distinct elements of $k, \neq 0, 1$.
- (d) Let Σ_6 be the symmetric group on 6 letters. Define an action of Σ_6 on sets of three distinct elements $\beta_1, \beta_2, \beta_3$ of $k, \neq 0, 1$, as follows: reorder the set $0, 1, \infty, \beta_1, \beta_2, \beta_3$ according to a given element $\sigma \in \Sigma_6$, then renormalize as in (c) so that the three become $0, 1, \infty$ again. Then the last three are the new $\beta'_1, \beta'_2, \beta'_3$.
- (e) Summing up, conclude that there is a one-to-one correspondence between the set of isomorphism classes of curves of genus 2 over k , and triples of distinct elements $\beta_1, \beta_2, \beta_3$ of $k, \neq 0, 1$, modulo the action of Σ_6 described in (d). In particular, there are many non-isomorphic curves of genus 2. We say that curves of genus 2 depend on three parameters, since they correspond to the points of an open subset of $\mathbb{A}_k^3$ modulo a finite group.

Proof.

- (a) By Hurwitz's theorem, the degree of the ramification divisor is computed to be

$$\deg R = \deg K_X - 2 \deg K_{\mathbb{P}^1} = 6.$$

The ramification index at a point cannot exceed the degree of the morphism, so f must be ramified at exactly 6 distinct points each having ramification index 2.

- (b) Let $B = k[x, z]/(z^2 - g(x))$ be an affine coordinate ring of X with $g(x) = (x - \alpha_1) \cdots (x - \alpha_6)$, and let f be the projection morphism onto the x -axis. By (II, 8.3A), we have

$$\Omega_{B/k} \cong (Bdx \oplus Bdz)/(g'(x)dx + 2zdz).$$

Let $P = (x', z')$ be a closed point in X . If $z' \neq 0$, then z is invertible in $\mathcal{O}_{P,X}$, so that

$$dz = -\frac{g'(x)}{2z}dx,$$

which means dx generates $(\Omega_{B/k})_P$. If $z' = 0$, then $x' = \alpha_i$ for some i , and $g'(x)$ is invertible in $\mathcal{O}_{P,X}$, so that

$$dx = -\frac{2z}{g'(x)}dz,$$

which means dz generates $(\Omega_{B/k})_P$. Since dx is a local parameter of $\mathbb{P}^1$, we see that P is ramified if and only if $z' = 0$ with ramification index $e_P = v_P(dx/dz) + 1 = 2$. There are six ramification points corresponding to the six roots of $g(x)$. Hence, $g(X) = 2$ by the Hurwitz formula. □

3. Plane Curves. Let X be a curve of degree d in $\mathbb{P}^2$. For each $P \in X$, let $T_P(X)$ be the tangent line to X at P (I, Ex.7.3). Considering $T_P(X)$ as a point of the dual projective plane $(\mathbb{P}^2)^*$, the map $P \rightarrow T_P(X)$ gives a morphism of X to its *dual curve* X^* in $(\mathbb{P}^2)^*$ (I, Ex. 7.3). Note that even though X is nonsingular, X^* in general will have singularities. We assume $\text{char } k = 0$ below.

- (a) Fix a line $L \subseteq \mathbb{P}^2$ which is not tangent to X . Define a morphism $\varphi : X \rightarrow L$ by $\varphi(P) = L \cap T_P(X)$, for each point $P \in X$. Show that φ is ramified at P if and only if either (1) $P \in L$, or (2) P is an *inflection point* of X , which means that the intersection multiplicity (I, Ex. 5.4) of $T_P(X)$ with X at P is ≥ 3 . Conclude that X has only finitely many inflection points.
- (b) A line of $\mathbb{P}^2$ is a *multiple tangent* of X if it is tangent to X at more than one point. It is a *bitangent* if it is tangent to X at exactly two points. If L is a multiple tangent of X , tangent to X at the points $P_1, \dots, P_r$, and if none of the P_i is an inflection point, show that the corresponding point of the dual curve X^* is an *ordinary r -fold point*, which means a point of multiplicity r with distinct tangent directions (I, Ex. 5.3). Conclude that X has only finitely many multiple tangents.
- (c) Let $O \in \mathbb{P}^2$ be a point which is not on X , nor on any inflectional or multiple tangent of X . Let L be a line not containing O . Let $\psi : X \rightarrow L$ be the morphism defined by projection from O . Show that ψ is ramified at $P \in X$ if and only if the line OP is tangent to X at P , and that the ramification index is 2. Use Hurwitz's theorem and (I, Ex. 7.2) to conclude that there are exactly $d(d-1)$ tangents of X passing through O . Hence, the degree of the dual curve (sometimes called the *class* of X) is $d(d-1)$.
- (d) Show that for all but a finite number of points of X , a point O of X lies on exactly $(d+1)(d-2)$ tangents of X , not counting the tangent at O .
- (e) Show that the degree of the morphism φ of (a) is $d(d-1)$. Conclude that if $d \geq 2$, then X has $3d(d-2)$ inflection points, properly counted. (If $T_P(X)$ has intersection multiplicity r with X at P , then P should be counted $r-2$ times as an inflection point. If $r = 3$, we called it an *ordinary inflection point*.) Show that an ordinary inflection point of X corresponds to an ordinary cusp of the dual curve X^* .
- (f) Now let X be a plane curve of degree $d \geq 2$, and assume that the dual curve X^* has only nodes and ordinary cusps as singularities (which should be true for sufficiently general X .) Then show that X has exactly $\frac{1}{2}d(d-2)(d-3)(d+3)$ bitangents.
- (g) For example, a plane cubic curve has exactly 9 inflection points, all ordinary. The line joining any two of them intersects the curve in a third one.
- (h) A plane quartic curve has exactly 28 bitangents. (This holds even if the curve has a tangent with four-fold contact, in which case the dual curve X^* has a tacnode.)

Proof.

- (a) Find local affine coordinates x, y of $\mathbb{P}^2$ such that

$$X : y - f(x) = 0, \quad L : x = 0.$$

Setting x and y to be local parameters for X at P and L at $\varphi(P)$, respectively, we can write φ in the form

$$\varphi(x) = f(x) - xf'(x).$$

It follows that

$$\varphi^* dy = \frac{dy}{dx} dx, \quad \frac{dy}{dx} = -xf''.$$

By (2.2), φ is ramified at P if and only if $v_P(dt/du) > 0$, where we have

$$v_P\left(\frac{dy}{dx}\right) = v_P(x) + v_P(f'').$$

Notice that $v_P(x) > 0$ if and only if $P \in L$. Also, it is not hard to see

$$\mu_P(X, T_P(X)) = v_P(f'') + 2.$$

Thus, $v_P(dt/du) > 0$ if and only if $v_P(x) > 0$ or $v_P(f'') > 0$ if and only if (1) $P \in L$ or (2) P is an inflection point of X . Since $\varphi : X \rightarrow L$ ramifies at finitely many points, X has finitely many inflection points.

- (b) Find local affine coordinates x, y of $\mathbb{P}^2$ such that

$$X : y - f(x) = 0, \quad L : y = 0.$$

This means x is a local parameter of X , so we can write $\gamma : X \rightarrow X^* \subset (\mathbb{P}^2)^*$ in the form

$$\gamma(x) = (-f'(x), xf'(x) - f(x)).$$

The tangent directions of $Q = \gamma(P_i)$ are precisely the images of the tangent vectors at P_i under the differential of γ . In particular, identifying the tangent vector of X at P_i to the row vector $(1, 0) \in T_{P_i}(X)$ and applying $d\gamma$ gives

$$v_i = (-f''(x_i), x_i f''(x_i)) \in T_Q(X^*).$$

We have shown in (a) that $f''(x_i) = 0$ if and only if P_i is an inflection point, which implies v_i and v_j define the same direction if and only if $P_i = P_j$. Hence, Q is an ordinary r -fold point with distinct tangent directions. Each multiple tangent of X corresponds to singular point of X^* , but a curve can have only finitely many singular points. Hence, X has finitely many multiple tangents.

- (c) Find local affine coordinates x, y of $\mathbb{P}^2$ such that

$$X : y - f(x) = 0, \quad L : x = 0, \quad O = (1 : 0 : 0).$$

We can set $u = x$ and $t = y$ to be local parameters for X at P and L at $Q = \psi(P)$, respectively, and ψ is given by the projection morphism onto the x -axis. It follows that

$$\psi^* dy = \frac{dy}{dx} dx, \quad \frac{dy}{dx} = f'.$$

Hence, ψ is ramified at $P \in X$ if and only if $f'(P) = 0$ if and only if the tangent of X at P passes through O . Suppose ψ is ramified at P . Since O is not on any inflectional or multiple tangents of X , $v_P(f'') = 0$. Hence, $v_P(f') = v_P(f'') + 1 = 1$.

The genus of X is related to d by the degree genus formula. Since ψ has degree d as a morphism of curves, Hurwitz's theorem can be used to compute the degree of the ramification divisor

$$\deg R = \deg K_X - d \deg K_{\mathbb{P}^1} = d(d-1).$$

The ramification index of each ramification point is 2, so the degree of R equals to the number of tangents passing through O .

- (d) This is another application of Hurwitz's theorem. The projection morphism $\psi : X - O \rightarrow L$ has degree $d-1$, so the degree of the ramification divisor of ψ is

$$\deg R = \deg K_X - (d-1) \deg K_{\mathbb{P}^1} = (d+1)(d-2).$$

A general point of X does not lie on any multiple tangents of X . Hence, the degree of R equals to the number of tangents passing through O .

- (e) Define $\pi : (\mathbb{P}^2)^* - L \rightarrow L$ by $L' \mapsto L \cap L'$. Since φ factors into $\varphi = \pi \circ \gamma$, the degree of φ equals to the degree of the image of γ , which is $d(d-1)$. The degree of the ramification divisor of φ is

$$\deg R = \deg K_X - d(d-1) \deg K_{\mathbb{P}^1} = 3d^2 - 5d.$$

The degree of the ramification index also counts the number of points $X \cap L$. In particular, $X \cap L$ is generically d distinct points, and the ramification index of each $P \in X \cap L$ is 2, so we need to subtract d from $\deg R$ to get the total number of inflection points of X , which equals to $3d(d-2)$. If $r = 3$, an ordinary inflection point of X corresponds to a twisted cubic in the dual curve locally, which has an ordinary cusp.

- (f) Let β_1 and β_2 be the number of bitangents and inflection points of X , respectively. We have two ways of computing the arithmetic genus of X^* : (1) in terms of $p_a(X)$, and (2) $X^* \subset (\mathbb{P}^2)^*$ as a plane curve of degree $d(d-1)$.

(1) The dual morphism $\gamma : X \rightarrow X^*$ defined in (b) is generically one-to-one by, so γ is a birational onto its image. The result of (Ex. 1.8) says

$$p_a(X^*) = p_a(X) + \delta, \quad \delta = \sum_{P \in X^*} \delta_P,$$

where $\delta_P = 1$ if $P \in X^*$ is a node or ordinary cusp, zero otherwise. In particular, $\delta = \beta_1 + \beta_2$ by (c) and (e).

(2) On the otherhand, X^* is a plane curve of degree $d(d-1)$ in $(\mathbb{P}^2)^*$, so the degree-genus formula for plane curves (III, Ex. 4.7) says

$$p_a(X^*) = \frac{1}{2}(d-2)(d+1)(d^2-d-1)$$

A smooth plane curve of degree d has $d(d-2)$ inflection points. Hence, the number of bitangents of X is

$$\beta_1 = p_a(X^*) - p_a(X) - \beta_2 = \frac{1}{2}d(d-2)(d-3)(d+3).$$

- (g) Let $E \subset \mathbb{P}^2$ be a plane cubic curve, let $P, Q \in E$ be two inflection points, and let L the line through P and Q . If $P = Q$, then L is the tangent line through P . In this case, P is an inflection point, so it is counted with a multiplicity of 3, i.e., L meets no other point of X . Otherwise, L cannot be a tangent line for the same reason, and L meets X at a distinct point $R \neq P, Q$. The morphism $\varphi : E \rightarrow L$ defined in (a) is ramified at R . By (e), φ has degree $= 3(3-1) = 6$. Since P and Q are ordinary inflection points, we have the ramification index of φ at P and Q is 2. Hence, R also has ramification index of 2, which is an inflection point.

□

4. *A Funny Curve in Characteristic p .* Let X be the plane quartic curve $x^3y + y^3z + z^3x = 0$ over a field of characteristic 3. Show that X is nonsingular, every point of X is an inflection point, the dual curve X^* is isomorphic to X , but the natural map $X \rightarrow X^*$ is purely inseparable.

Proof. The partial derivatives of $f = x^3y + y^3z + z^3x$ are

$$\partial_x f = z^3, \quad \partial_y f = x^3, \quad \partial_z f = y^3,$$

so in coordinates, the dual morphism $X \rightarrow X^*$ is given by

$$(x : y : z) \mapsto (x^3 : y^3 : z^3),$$

This immediately shows X is nonsingular, every point of X is an inflection point, and the natural map $X \rightarrow X^*$ is purely inseparable. Also, we can identify $X^* \subset (\mathbb{P}^2)^*$ with the image of X under the k -linear Frobenius morphism $\mathbb{P}^2 \rightarrow \mathbb{P}^2$ (2.4.1), and under this identification, $f^3 = 0$ is a defining equation of X^* . Hence, X and X^* are isomorphic as abstract projective k -schemes. □

5. *Automorphisms of a Curve of Genus ≥ 2 .* Prove the theorem of Hurwitz [1] that a curve X of genus $g \geq 2$ over a field of characteristic 0 has at most $84(g-1)$ automorphisms. We will see later (Ex. 5.2) or (V, Ex. 1.11) that the group $G = \text{Aut } X$ is finite. So let G have order n . Then G acts on the function field $K(X)$. Let L be the fixed field. Then the field extension $L \subseteq K(X)$ corresponds to a finite morphism of curves $f : X \rightarrow Y$ degree n .

- (a) If $P \in X$ is a ramification point, and $e_P = r$, show that $f^{-1}f(P)$ consists of exactly n/r points, each having ramification index r . Let $P_1, \dots, P_s$ be a maximal set of ramification points of X lying over distinct points of Y , and let $e_{P_i} = r_i$. Then show that Hurwitz's theorem implies that

$$(2g - 2)/n = 2g(Y) - 2 + \sum_{i=1}^s (1 - 1/r_i).$$

- (b) Since $g \geq 2$, the left hand side of the equation is > 0 . Show that if $g(Y) \geq 0$, $s \geq 0$, $r_i \geq 2$, $i = 1, \dots, s$ are integers such that

$$2g(Y) - 2 + \sum_{i=1}^s (1 - 1/r_i) > 0,$$

then the minimum value of this expression is $1/42$. Conclude that $n \leq 84(g - 1)$.

Proof.

- (a) Each $\sigma \in G$ induces an automorphism of X such that $f \circ \sigma = f$, so there is a group action of G on each fibre $f^{-1}f(P)$. The action is transitive (A.M., Ex. 5.13), and $e_P = e_{\sigma(P)}$ because σ induces an isomorphism of local rings $\mathcal{O}_{\sigma(P_i), X} \rightarrow \mathcal{O}_{P_i, X}$, hence sends uniformizer to uniformizer. In particular, if d_i is the size of the orbit of P_i , then $n = d_i r_i$. Hence, we have

$$\sum_{P \in X} (e_P - 1) = \sum_{i=1}^s \sum_{P \in f^{-1}f(P_i)} (e_P - 1) = \sum_{i=1}^s d_i (r_i - 1) = n \sum_{i=1}^s (1 - 1/r_i).$$

- (b) The following gives the minimum value

$$g(Y) = 0, \quad s = 3, \quad r_1 = \frac{1}{2}, \quad r_2 = \frac{1}{3}, \quad r_3 = \frac{1}{7}.$$

We conclude that

$$n \leq 42(2g - 2) = 84(g - 1).$$

□

- 6. f_* for Divisors.** Let $f : X \rightarrow Y$ be a finite morphism of curves of degree n . We define a homomorphism $f_* : \text{Div } X \rightarrow \text{Div } Y$ by $f_*(\sum n_i P_i) = \sum n_i f(P_i)$ for any divisor $D = \sum n_i P_i$ on X .

- (a) For any locally free sheaf $\mathcal{E}$ on Y , of rank r , we define $\det \mathcal{E} = \bigwedge^r \mathcal{E} \in \text{Pic } Y$ (II, Ex. 6.11). In particular, for any invertible sheaf $\mathcal{M}$ on X , $f_* \mathcal{M}$ is locally free of rank n on Y , so we can consider $\det f_* \mathcal{M} \in \text{Pic } Y$. Show that for any divisor D on X ,

$$\det(f_* \mathcal{O}_X(D)) \cong (\det f_* \mathcal{O}_X) \otimes \mathcal{O}_Y(f_* D).$$

Note in particular that $\det(f_* \mathcal{O}_X(D)) \neq \mathcal{O}_Y(f_* D)$ in general!.

- (b) Conclude that $f_* D$ depends only on the linear equivalence class of D , so there is an induced homomorphism $f_* : \text{Pic } X \rightarrow \text{Pic } Y$. Show that $f_* \circ f^* : \text{Pic } Y \rightarrow \text{Pic } Y$ is just multiplication by n .
- (c) Use duality for a finite flat morphisms (III, Ex. 6.10) and (III, Ex. 7.2) to show that

$$\det f_* \Omega_X \cong (\det f_* \mathcal{O}_X)^{-1} \otimes \Omega_Y^{\otimes n}.$$

- (d) Now assume that f is separable, so we have the ramification divisor R . We define the *branch divisor* B to be the divisor $f_* R$ on Y . Show that

$$(\det f_* \mathcal{O}_X)^2 \cong \mathcal{O}_Y(-B).$$

Proof.

- (a) Let P be any point in X , and let $f_*P = f(P)$ be the image of P . The skyscraper sheaf $k(f_*P)$ has a natural free resolution

$$0 \longrightarrow \mathcal{O}_Y(-f_*P) \longrightarrow \mathcal{O}_Y \longrightarrow k(f_*P) \longrightarrow 0.$$

In particular, $\det k(f_*P) \cong \mathcal{O}_Y(f_*P)$, and any finite free resolution of $k(f_*P)$ induces isomorphic determinant line bundles (II, Ex. 6.11). Also, the sheaves $k(f_*P) \rightarrow f_*k(P)$ are isomorphic, so any free resolution of $f_*k(P)$ is also a free resolution of $k(f_*P)$.

Now, let D be any non-zero divisor, and let P be any point. We will show that the formula is true for D if and only if it is true for $D + P$ (1.3). Consider the exact sequence

$$0 \longrightarrow \mathcal{O}_X(-P) \longrightarrow \mathcal{O}_X \longrightarrow k(P) \longrightarrow 0.$$

Tensoring with $\mathcal{O}_X(D + P)$ and applying f_* gives

$$0 \longrightarrow f_*\mathcal{O}_X(D) \longrightarrow f_*\mathcal{O}_X(D + P) \longrightarrow f_*k(P) \longrightarrow 0.$$

By the remarks above, we have an isomorphism

$$\mathcal{O}_Y(f_*P) \cong \det(f_*\mathcal{O}_X(D + P)) \otimes (\det(f_*\mathcal{O}_X(D)))^{-1},$$

which is what we wanted to show.

- (b) By (a), since $f_*\mathcal{O}_X(D)$ depends only on the linear equivalence class of D , so does $\mathcal{O}_Y(f_*D)$. For the second part, it suffices to show the statement holds for any point $Q \in Y$. Indeed, it follows by definition $f_* \circ f^*$ is multiplication by some integer. On the other hand, f^* multiplies the degree by n (II, 6.9), and f_* preserves degrees. Hence, $f_* \circ f^*$ must be multiplication by n .
- (c) Let $\mathcal{F}$ be a coherent sheaf on Y . By definition, we have

$$f_*f^!\mathcal{F} \cong f_*\mathcal{O}_X^{-1} \otimes \mathcal{F}, \quad f_*\mathcal{O}_X^{-1} = \mathcal{H}om_Y(f_*\mathcal{O}_X, \mathcal{O}_Y).$$

By (III, Ex. 7.3), $f^!\Omega_Y \cong \Omega_X$ is a dualizing sheaf on X . Applying above to $\mathcal{F} = \Omega_Y$ gives

$$\begin{aligned} \det f_*\Omega_X &\cong \det f_*f^!\Omega_Y \\ &\cong \det(f_*\mathcal{O}_X^{-1} \otimes \Omega_Y) \\ &\cong (\det f_*\mathcal{O}_X)^{-1} \otimes \Omega_Y^{\otimes n}. \end{aligned}$$

- (d) Let K_X and K_Y be canonical divisors of X and Y , respectively. They satisfy the following relation (2.3)

$$f^*K_Y - K_X \sim -R.$$

Applying f_* gives

$$nK_Y - f_*K_X \sim -B.$$

On the otherhand, (a) applied to $D = K_X$ gives

$$\mathcal{O}_Y(-f_*K_X) \cong (\det f_*\mathcal{O}_X) \otimes (\det f_*\Omega_X)^{-1}.$$

and by (c), we have the isomorphism

$$\mathcal{O}_Y(nK_Y) \cong (\det f_*\mathcal{O}_X) \otimes (\det f_*\Omega_X),$$

Combining these two isomorphisms gives

$$(\det f_*\mathcal{O}_X)^2 \cong \mathcal{O}_Y(nK_Y - f_*K_X) \cong \mathcal{O}_Y(-B).$$

□

- 7. Étale Covers of Degree 2.** Let Y be a curve over a field k of characteristic $\neq 2$. We show there is a one-to-one correspondence between finite étale morphisms $f : X \rightarrow Y$ of degree 2, and 2-torsion elements of $\text{Pic } Y$, i.e., invertible sheaves $\mathcal{L}$ on Y with $\mathcal{L}^2 \cong \mathcal{O}_Y$.

- (a) Given an étale morphism $f : X \rightarrow Y$ of degree 2, there is a natural map $\mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$. Let $\mathcal{L}$ be the cokernel. Then $\mathcal{L}$ is an invertible sheaf on Y , $\mathcal{L} \cong \det f_*\mathcal{O}_X$, and so $\mathcal{L}^2 \cong \mathcal{O}_X$ by (Ex. 2.6). Thus an étale cover of degree 2 determines a 2-torsion element in $\text{Pic } Y$.
- (b) Conversely, given a 2-torsion element $\mathcal{L}$ in $\text{Pic } Y$, define an $\mathcal{O}_Y$ -algebra structure on $\mathcal{O}_Y \oplus \mathcal{L}$ by $\langle a, b \rangle \cdot \langle a', b' \rangle = \langle aa' + \varphi(b \otimes b'), ab' + a'b \rangle$, where φ is an isomorphism of $\mathcal{L} \otimes \mathcal{L} \rightarrow \mathcal{O}_Y$. Then take $X = \text{Spec}(\mathcal{O}_Y \oplus \mathcal{L})$. Show that X is an étale cover of Y .
- (c) Show that these two processes are inverse to each other.

Proof.

- (a) We check that $\mathcal{L}$ is an invertible sheaf on Y . At the level of stalks, the natural map $\mathcal{O}_{Q,Y} \rightarrow (f_*\mathcal{O}_X)_Q \cong \mathcal{O}_{Q,Y}^{\oplus 2}$ is the diagonal map, so the cokernel is free of rank one. Hence, $\mathcal{L}$ is locally free of rank one.
- (b) By construction, $f : X \rightarrow Y$ is a finite morphism, and $f_*\mathcal{O}_X \cong \mathcal{O}_Y \oplus \mathcal{L}$ is locally free of rank 2, so f is a flat morphism. Also, φ is locally given by multiplication, i.e., $\varphi(b \otimes b') = bb'$. If P is a point in Y , then there is an isomorphism

$$(f_*\mathcal{O}_X)_P \cong \mathcal{O}_{P,X}[\varepsilon]/(\varepsilon^2 - 1) \cong \mathcal{O}_{P,X} \times \mathcal{O}_{P,X},$$

so f is unramified at every $P \in Y$. Hence, $f : X \rightarrow Y$ is a degree 2 étale cover of Y .

- (c) One direction is obvious. Consider the pullback diagram

$$\begin{array}{ccc} X \times_Y X & \longrightarrow & X \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}.$$

The projection morphisms $p_1, p_2 : X \times_Y X \rightarrow X$ are étale covers of degree 2. Also, the diagonal $\Delta \subset X \times_Y X$ is a closed subscheme of $X \times_Y X$ isomorphic to X . Thus, the restriction $p_1, p_2 : X \times_Y X - \Delta \rightarrow X$ is an étale cover of X of degree 1, which is an isomorphism. Hence, $\tau = p_2 \circ p_1^{-1} : X \rightarrow X$ is an automorphism of X over Y which transposes the fibres of f .

Now, let $f : X \rightarrow Y$ be a étale cove of degree 2, and let $X' = \text{Spec}(\mathcal{O}_Y \oplus \mathcal{L})$, where $\mathcal{L}$ is the cokernel of the natural morphism $\mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$. We want to show X and X' are isomorphic. It suffices to show the following short exact sequence of $\mathcal{O}_Y$ -modules splits

$$0 \longrightarrow \mathcal{O}_Y \longrightarrow f_*\mathcal{O}_X \longrightarrow \mathcal{L} \longrightarrow 0.$$

The automorphism $\tau : X \rightarrow X$ of X over Y , which transposes the fibres of f , induces an $\mathcal{O}_Y$ -module automorphism $\tau_* : f_*\mathcal{O}_X \rightarrow f_*\mathcal{O}_X$. The kernel of the trace morphism $t = 1 - \tau : f_*\mathcal{O}_X \rightarrow f_*\mathcal{O}_X$ is exactly $\mathcal{O}_Y$. Hence, t induces a section $\mathcal{L} \rightarrow f_*\mathcal{O}_X$, so the short exact sequence splits. $\square$